**AAE 301 Signal Analysis for Aerospace Engineering**
**3 credits**

## 1 Course Information/Description

Signal processing and spectral analysis for aerospace engineering. Fourier and fast Fourier transforms. Vibration analysis; estimation of natural frequencies, wing vibrations. Introduction to linear circuits, operational amplifiers and filtering. Noise suppression.

***Brightspace Learning Management System:***

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

***Textbook:***

We will be based on source notes and slides.

## 2 Assignments

The course has individual assignments. Homework will be assigned on Friday by 5 PM ET and due on Friday at 11:59 PM ET via Gradescope.

## 3 Course Learning Objectives

1. Use fast Fourier transform techniques to estimate sinusoids from experimental data.
2. Estimate natural frequencies from data. For example, beam vibration.
3. Extract the underlying spectrum from time history.
4. An elementary understanding of stability and state space systems.
5. An elementary understanding of transfer functions and Bode plots.
6. Analyze mass spring damper systems.
7. Analyze and design elementary linear circuits.
8. Design low pass, band pass and high pass filters.

## 4 Evaluation

All assignments, presentations, etc., must be submitted on BS or Gradescope as instructed. No email submissions will be accepted. Late submissions will be accepted only in exceptional circumstances. Make sure to communicate with me clearly and ASAP in case you anticipate a late submission, or have missed a submission.

Your grade will be calculated as follows:

| Assignments | Weight |
|---|---|
| Quizzes | 15% |
| Homework | 25% |
| Midterm Exam | 25% |
| Final Exam | 35% |

Your course grade will be based on the following grading scale:

| Letter Grade | Percentage |
|---|---|
| A+ | 97-100% |
| A | 92-96% |
| A- | 90-91% |
| B+ | 87-89% |
| B | 84-86% |
| B- | 80-83% |
| C+ | 77-79% |
| C | 74-76% |
| C- | 70-73% |

| Letter Grade | Percentage |
|---|---|
| D+ | 67-69% |
| D | 64-66% |
| D- | 60-63% |
| F | 0-59% |